The Pascaline

West bay 2: Pascal's adding machine (1642) · 14 November 2026

Babbage's Hall, west bay 2. Blaise Pascal's adding machine (1642), with three digits and his own carry: the **sautoir** ("jumper").

Each digit wheel carries ten pins and, behind them, a snail cam; in front of the plate, its dial shows the digit. As the units count from 0 to 9 the cam slowly lifts a weighted lever, the sautoir. As the units pass 9 the cam's step drops away under it, the weight falls, and a pawl on the lever's end kicks the tens wheel on by one pin. On the way up the pawl yields and slips back over the next pin. Because each carry is powered by a weight that was lifted earlier, a carry that ripples (99 + 1 = 100) needs no more force from the hand than any other step. This was Pascal's answer to the problem that wrecked earlier calculators: a long ripple carry jams a machine that drives it straight through gears.

Using it

As with Pascal's own, by hand, with a stylus. Each dial has ten holes between its numbers, and round each dial the front plate has the numbers 1 to 9, with a finger stop at about 4 o'clock. To add 7 to a digit, put the stylus in the dial's hole beside the plate's 7 and turn it clockwise until it meets the stop: the digit goes on by exactly 7, whatever it showed. To add 84, turn the units by 4 and the tens by 8. The total is read under the pointers. A digit passing 9 carries one into the next by itself, on the weight it lifted on the way.

It only goes forward (turned back, the sautoir's nose would meet the steep side of the cam). Pascal subtracted by adding the complement: to take 3 from a three-digit number, add 997 and let the carry out of the hundreds fall away.

In the simulation you drive it the same way, with the mouse or the keys:

- **The mouse:** rest it on a hole of any dial. The stylus goes into that hole, and the tooltip says what it will add ("add 7 to the units: 000 becomes 007"). Press **Space** and it turns to the stop. Nothing turns until you press Space, so the mouse can wander freely. (bpp asks about the mouse once it has been still for half a second.)
- **The keys:** `U`, `T` or `H` chooses the units, tens or hundreds (a brass mark appears below that dial); `1` to `9` adds that much to it, in one stroke: the stylus goes into the hole by the plate's number and round to the stop.

The hand that turns the digit is a motor on its axle; the stylus is drawn doing it, and lies on the table in front between strokes.

Running it

Open `pascaline.lua` in bpp, or run it inside the hall (`HALL_START=pascaline`). It starts with a tour: 7, then + 8 = 15 (a carry), + 84 = 99, + 1 = 100 (a double ripple). It prints `RESULT pascaline tour: PASS` when every reading is right.

Key	
`U` `T` `H`	choose the units, tens or hundreds

Key	
`1` to `9`	add that much to the chosen wheel
`Space`	turn the stylus in the hole the mouse chose to the stop
`R`	reset to 000
`D`	the tour again
`Z`	slow motion (1/4 speed) on and off

Sliders: `friction` (0.15 to 0.45 checked), `turn rate` (how fast the hand turns a digit), `speed`.

Environment, for testing: `PA_TOUR=0` (no tour), `PA_FRICTION`, `PA_RATE`, `PA_DEBUG=1` (prints the digits' values and the levers' and pawls' angles), `PA_EXPORT=1` (rewrites `parts/geometry.scad`), `PA_REST=0` (never rest: see below), `PA_REST_LOG=1` (says when it rests), `PA_DET_T_MM` (the detents' leaves this thick, in mm, to try others).

Once the hand is done and nothing moves, the machine rests: it stops stepping until the next key or tour step, which saves the processor and changes nothing it computes.

How it is built

- **One source.** `pascaline-geometry.lua` holds every moving part's shape as boxes, cylinders and convex hulls, with masses. The simulation builds its bodies from them, and writes them to `parts/geometry.scad`, which `parts/pascaline.scad` builds the printed parts from. What is printed is what was simulated.
- **Bodies.** Each lever and pawl is a compound of its primitives, its origin at its centre of mass. The digits turn on hinges to the frame; the sautoirs pivot on pins in the back plate, at the axles' height; each pawl is hinged to its sautoir, limited so it yields one way only, and held against its stop by a small tail weight.
- **The detents.** Each wheel is held at its digit by a printed PLA leaf spring, whose round nose rests between two pins at 9 o'clock and centres the wheel. It is simulated as a body (nose and leaf) on a hinge at the leaf's root, turned back by the leaf's own stiffness, worked out from PLA (3EI/L), with the 0.5 mm preload it has in place.
- **The hand** turns one digit at a time by its hinge's motor, one step per press, then lets go.

Printing and building

`./make-stls.sh` (needs OpenSCAD, and bpp to refresh `geometry.scad`) writes every part into `stl/`, each turned to lie on the bed. The largest pieces are 230 x 140 mm.

Part	Print	Size (mm)	Notes
`digit-cam.stl`	2	72 × 83 × 43	units and tens. Cam on the bed; supports under the pin wheel
`digit-last.stl`	1	59 × 59 × 32	hundreds (no cam)
`dial.stl`	3	88 × 88 × 6	press onto the hub's end. Print it light (ivory or white) and paint the raised numbers dark
`sautoir-0.stl`, `sautoir-1.stl`	1 each	215 × 28 x 40/52	supports under the pawl's carrier arm
`pawl-0.stl`, `pawl-1.stl`	1 each	31 × 23 × 8	
`stop.stl`	3	13 × 32 × 5	the finger stops: printed on their side, screwed to the front plate

Part	Print	Size (mm)	Notes
`stylus.stl`	1	8 × 8 × 30	or use any 4 mm rod
`detent.stl`	3	19 × 55 × 14	the leaf springs. PLA, standing as it lies (no supports), 100% infill; the leaf is 1.2 mm, three lines of a 0.4 mm nozzle
`frame-front-left/right.stl`	1 each	230 × 140 x 15	the front plate, the digits' bearings, the pointers and the numbers round each dial (paint them)
`frame-back-left/right.stl`	1 each	230 × 140 x 6	the back plate, the sautoirs' pivots
`frame-base-left/right.stl`	1 each	230 × 110 x 8	
`frame-end.stl`	2	58 × 132 × 8	joins the plates at each end

Instead of printing the plates you can cut them from 6 mm acrylic (clear, so the works show) or plywood, gluing on three printed `pointer.stl`: `stl/cut-front-plate.svg` and `stl/cut-back-plate.svg` are their outlines.

Also needed

- 2 steel weights of about **60 g**, each on an **M6 bolt** through the sautoir's weight block (a stack of M6 nuts or washers does). The weight matters: the drop must carry the next wheel a full step.
- 2 pawl tail weights of about **8 g** (an M5 nut glued on). Too light and the pawl swings out instead of catching.
- **3 mm steel rod** for pivot pins: 2 for the sautoirs (into the back plate) and 2 for the pawls.
- **3 M3 screws,** about 16 mm, one through the front plate into each detent's block (its 2.5 mm hole is for the screw to tap).
- **3 M3 screws,** about 8 mm, one through each finger stop into the front plate (tapped into its 2.5 mm hole).
- Glue or M3 screws for the frame joints; a little dry PTFE or graphite on the cams and pins.

Assembly. Glue the plate halves and the base halves. Fit the sautoirs on their pins in the back plate, and the pawls on theirs. Slide each digit's hub, cam first, from behind through the front plate, then press its dial on with 0 at the top under the pointer (its holes then stand by the plate's numbers). Set the units and tens to 0 with each sautoir resting at the bottom of its cam. Screw a detent behind each wheel, its block against the back of the front plate and its nose dropped between the two pins at 9 o'clock (lift the nose over a pin to fit it). Join the plates with the end pieces and the base. Screw a finger stop beside each dial, its arm reaching in over the dial's face. Then add with the stylus (see "Using it") and watch the tens step on as the units pass 9.

The detents' thickness matters: the simulation's carry works with leaves from 1.0 to 1.6 mm thick (see `FINDINGS.md`), and 1.2 mm is in the middle. If your printer makes them thicker or thinner, the carry still has room.

See `FINDINGS.md` for how the numbers were arrived at.